Ayush Anmol

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Education

International Institute of Information Technology (IIIT) Bhubaneswar, India
Bachelor of Technology in Computer Science and Engineering *Nov. 2022 – Jun. 2026*

Work Experience

Summer Intern | LangChain, Google ADK, RAG, OCR May 2025 – Jul. 2025
Neophyte Ambient Intelligence Private Limited (Neophyte.ai) *Remote*

- Designed and implemented a real-time Speech-to-Speech system with a latency of 0.4s.
- Developed RAG models for data processing and information retrieval achieving an accuracy of 0.90.
- Automated text recognition for 110 scanned invoices by integrating an OCR model, reducing manual efforts.

Project Intern | LangChain, Streamlit, CSS, Ollama Dec. 2024 – Jan. 2025
Hindustan Petroleum Corporation Limited (HPCL) *Mumbai, India*

- Engineered a conversational database agent with automated query refinement and schema analysis tools.
- Built a web interface for real-time data display (latency ∼3s), with session memory and interactive user features.
- Improved workflow efficiency with a modular UI–agent–database design, validated on 100+ queries.

Research Intern | Scikit-learn, PyTorch, QGIS May 2024 – Aug. 2024
Indian Space Research Organization (ISRO) *Patna, India*

- Processed and analyzed over 350 multi-spectral satellite images for land cover classification and segmentation.
- Trained a R-CNN in PyTorch to analyze and classify satellite images for land-use and land-cover mapping.
- Serves as a potential prototype for ISRO's 5 year live project to enhance land cover classification by up to ∼5%.

Projects

Semantic Book Recommender | *Python, Ollama, LangChain, Gradio*

- Scope: A semantic recommender system to suggest books beyond keyword matching using embedding similarity.
- Operation: Integrated embedding model (Ollama) with LangChain retrieval pipelines, deployed using Gradio and tested on 6000+ book entries.
- Impact: Showcased measurable gains in recommendation quality through embedding-based retrieval.

Medical Chatbot | *Python, LangChain, FAISS, Streamlit*

- Scope: An interactive chatbot to assist patients with accurate medical Q&A using RAG.
- Operation: Integrated HuggingFace embeddings with FAISS vector search and GPT-based generation; tested on 1000+ queries with 92% retrieval accuracy and ∼1.5s latency.
- Impact: Improved patient accessibility to reliable medical info, demonstrating AI's role in supporting healthcare.

Histopathological Image Classification | *Python, TensorFlow, OpenCV*

- Scope: A CNN-based model to classify histopathology images into cancerous vs. non-cancerous tissue.
- Operation: Trained on the Kaggle Histopathologic Cancer Detection dataset (300K+ images) with augmentation and preprocessing for robustness.
- Impact: Achieved 0.88 accuracy, showing potential for faster, AI-assisted diagnostics in digital pathology.

Technical Skills

Languages: Python, C/C++, SQL
DSA: Solved 200+ DSA problems on LeetCode (Rating: 1600+) (leetcode.com/u/Ayush114)
Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, PyTorch
ML Techniques: Linear & Logistic Regression, SVM, Decision Trees, NLP, Deep Learning (CNN, RNN, LSTM)
Tools & Frameworks: Streamlit, FastAPI, LangChain, LangGraph, Google ADK, Git, GitHub
Certifications: Machine Learning Specialization (Coursera)

Publications

Hierarchical BiLSTM Encoding for Vocal-Driven Cover Song Detection (OCIT-2025, IEEE)

Achievements

Selected for **Amazon ML Summer school 2025** (∼2-5% acceptance)
Finalist in **Space India Hackathon IISF 2023** (Top 5%)